

DELTA WEB DEVELOPMENT CAPSTONE

Full-Stack Accommodation Booking Platform

 Vaibhav
Patel

 B.Tech CSE
(AI/ML)

 IILM
University

Section: 2CSE13 | Roll No: 25scs1003003209

Project Overview

The Objective

Engineer a scalable, dynamic web application mirroring the core functionalities of industry-leading travel platforms (like Airbnb).

The Context

Developed independently from June to September 2026 under the Apna College "Delta" certification program.

The Result

A production-ready platform featuring secure authentication, dynamic user-generated content, and robust cloud media storage.



[Home](#) [All Listings](#) [Add new Listing](#)

All Listings



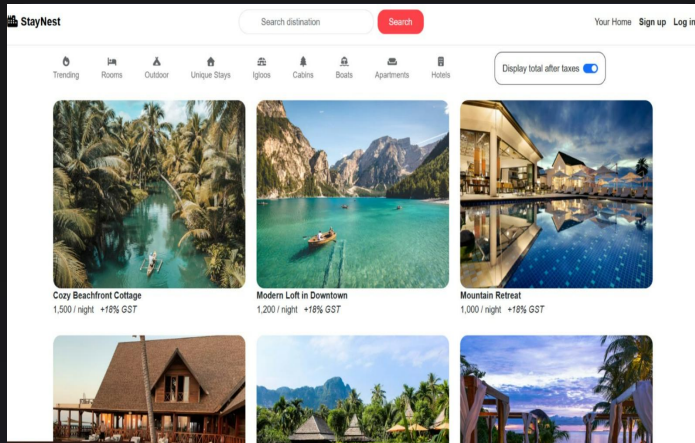
Cozy Beachfront Cottage
1,500 / night



Modern Loft in Downtown
1,200 / night

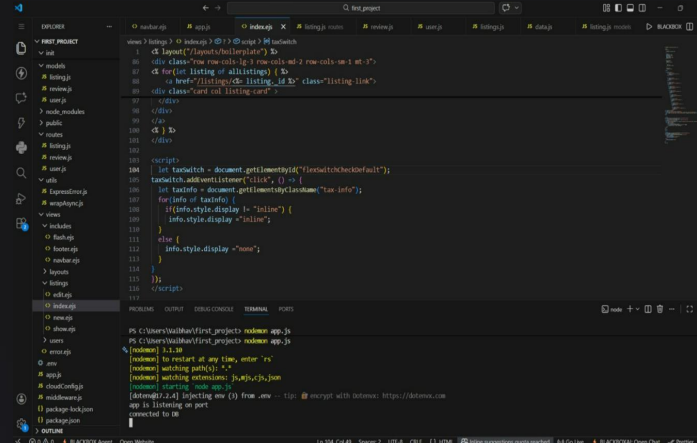


The Technology Stack M.E.R.N Stack



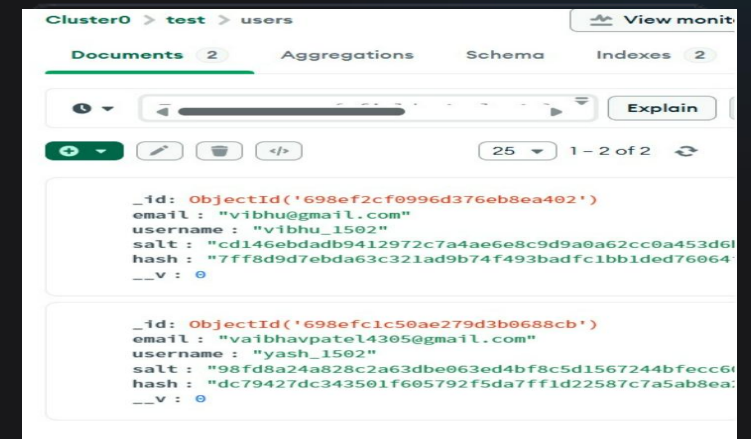
Frontend

HTML5, CSS3, Bootstrap 5 for responsive design, and EJS (Embedded JavaScript) for dynamic template rendering.



Backend

Node.js runtime environment with the Express.js framework for robust HTTP server creation and RESTful API routing.



Database

MongoDB NoSQL database managed through the Mongoose ODM, enabling strict data schemas and complex relational queries.

Core Features Capabilities



Complete CRUD

Users can seamlessly Create, Read, Update, and Delete custom property listings with dynamic form validations and responsive UI updates.



Secure Auth

Integrated session-based user login and registration with strict role-based access control (ensuring only owners can edit properties).



Interactive Reviews

Authenticated users can leave 1-5 star ratings and written reviews on properties, establishing a trust-based ecosystem.

System Architecture (MVC) *Data Flow*



Models (Data Layer) Defined strict, scalable database schemas for Listings, Reviews, and Users utilizing Mongoose object data modeling.



Views (Presentation Layer) Created modular EJS templates linked to a standardized boilerplate layout, ensuring dynamic and seamless frontend data rendering.



Controllers (Logic Layer) Engineered Express routing that intercepts HTTP requests, interacts with the MongoDB database, applies middleware logic, and serves the UI.

Key Learning Outcomes Skills Acquired



Backend Architecture Mastered the lifecycle of HTTP requests and designed efficient RESTful APIs from scratch using Express router.



Data Modeling Learned to manage relational data models within a NoSQL environment, specifically embedding review object IDs within property schemas.



Security Implementation Gained hands-on experience using Passport.js to secure routes, hash passwords, and safely manage environment variables (.env).



Error Handling & Validation Implemented robust Joi server-side validation and custom error-catching middleware to prevent application crashes and improve UX.

Official Certification Validation

Delta Program Completion

The rigorous training and project development culminated in the successful awarding of the "Delta (Full Stack Web Development)"

certificate by Apna College.
This certification serves as a formal validation of the technical competencies acquired during the 14-week program and the successful completion of the StayScape capstone project.

✓ Verified Full Stack Developer



Conclusion

Successfully bridged the gap between academic theory and practical software engineering. By delivering a fully functional, complex application, I established a strong, proven foundation in modern JavaScript frameworks and scalable backend infrastructure.